浙江大学2018-19秋冬《数据结构基础》期中模拟练习

开始时间 01/01/2016 8:00:00 AM **结束时间** 01/18/2038 8:00:00 AM **答题时长** 90分钟
考生 abl1h **得分** 3 **总分** 100

判断题

得分：3 总分：15

1-1 For a sequentially stored linear list of length N, the time complexities for query and insertion are $O(1)$ and $O(N)$, respectively. (3分)

(●) T ( ) F

评测结果 答案正确（3 分）

1-2 If keys are pushed onto a stack in the order {1, 2, 3, 4, 5}, then it is impossible to obtain the output sequence {3, 4, 1, 2, 5}. (3分)

( ) T ( ) F

评测结果 未作答（0 分）

1-3 2^N and N^N have the same speed of growth. (3分)

( ) T ( ) F

评测结果 未作答（0 分）

1-4 If there are less than 20 inversions in an integer array, then Insertion Sort will be the best method among Quick Sort, Heap Sort and Insertion Sort. (3分)

( ) T ( ) F

评测结果 未作答（0 分）

1-5 In a binary search tree, the keys on the same level from left to right must be in sorted (non-decreasing) order. (3分)

( ) T ( ) F

评测结果 未作答（0 分）

单选题

得分：暂无 总分：64

2-1 The result of performing three DeleteMin operations in the min-heap {1,3,2,6,7,5,4,15,14,12,9,10,11,13,8} is: (5分)

( ) A. 4,5,6,7,8,9,10,11,12,13,14,15

B. 4,6,5,13,7,10,8,15,14,12,9,11
C. 4,6,5,12,7,10,8,15,14,9,13,11
D. 4,5,6,12,7,10,8,15,14,13,9,11

2-2 Among the following sorting methods, which ones will be slowed down if we store the elements in a linked structure instead of a sequential structure? (5分)

1. Insertion sort; 2. Selection Sort; 3. Bubble sort; 4. Shell sort; 5. Heap sort

A. 1 and 2 only
B. 2 and 3 only
C. 3 and 4 only
D. 4 and 5 only

2-3 To delete `p` from a doubly linked list, we must do: (5分)

A. `p->prior=p->prior->prior; p->prior->next=p;`
B. `p->next->prior=p; p->next=p->next->next;`
C. `p->prior->next=p->next; p->next->prior=p->prior;`
D. `p->next=p->prior->prior; p->prior=p->next->next;`

2-4 Insert { 5, 11, 13, 1, 3, 6 } one by one into an initially empty binary search tree. The post-order traversal sequence of the resulting tree is: (5分)

A. 3, 1, 5, 6, 13, 11
B. 3, 1, 6, 13, 11, 5
C. 1, 3, 11, 6, 13, 5
D. 1, 3, 5, 6, 13, 11

2-5 Suppose that an array of size `m` is used to store a circular queue. If the front position is `front` and the current size is `size`, then the rear element must be at: (5分)

A. `front+size`
B. `front+size-1`
C. `(front+size)%m`
D. `(front+size-1)%m`

2-6 For an in-order threaded binary tree, if the pre-order and in-order traversal sequences are `B E A C F D` and `A E C B D F` respectively, which pair of nodes' right links are both threads? (4分)

A. B and E
B. E and F
C. A and E
D. A and D

2-7 The recurrent equations for the time complexities of programs P1 and P2 are:

- P1: $T(1)=1, T(N)=T(N/2)+1$;
- P2: $T(1)=1, T(N)=2T(N/2)+1$;

Then the correct conclusion about their time complexities is: (5分)

- A. they are both $O(logN)$
- B. $O(logN)$ for P1, and $O(N)$ for P2
- C. they are both $O(N)$
- D. $O(logN)$ for P1, and $O(NlogN)$ for P2

2-8 Given a quadtree(四叉树) with 4 nodes of degree 2, 4 nodes of degree 3, 3 nodes of degree 4. The number of leaf nodes in this tree is __. (5分)

- A. 22
- B. 21
- C. 20
- D. 12

2-9 How many leaf node does a complete binary tree with 2435 nodes have? (5分)

- A. 1218
- B. 1217
- C. 812
- D. cannot be determined

2-10 In-order traversal of a binary tree can be done iteratively. Given the stack operation sequence as the following:

```
push(1), push(2), push(3), pop(), push(4), pop(), pop(), push(5), pop(), pop(), push(6), p
```

Which one of the following statements is TRUE? (5分)

- A. 6 is the root
- B. 2 is the parent of 4
- C. 2 and 6 are siblings
- D. None of the above

2-11 To sort { 8, 3, 9, 11, 2, 1, 4, 7, 5, 10, 6 } by Shell Sort, if we obtain (4, 2, 1, 8, 3, 5, 10, 6, 9, 11, 7) after the first run, and (1, 2, 3, 5, 4, 6, 7, 8, 9, 11, 10) after the second run, then the increments of these two runs must be __ , respectively. (5分)

- A. 3 and 1
- B. 3 and 2
- C. 5 and 2
- D. 5 and 3

2-12 Given input { 4321, 56, 57, 46, 28, 7, 331, 33, 234, 63 }. Which one of the following is the result after the 1st run of the Least Signification Digit (LSD) radix sort? (5分)

- A. →331→4321→33→63→234→56→46→57→7→28
- B. →4321→331→33→63→234→56→46→57→7→28
- C. →56→28→4321→331→33→234→46→57→63→7
- D. →57→46→28→7→33→234→63→56→4321→331

2-13 For the quicksort implementation with the left pointer stops at an element with the same key as the pivot during the partitioning, but the right pointer does not stop in a similar case, what is the running time when all keys are equal? (5分)

- A. $O(logN)$
- B. $O(N)$
- C. $O(NlogN)$
- D. $O(N^2)$

程序填空题

得分：暂无　总分：15

5-1 The function is to sort the list { `r[1] … r[n]` } in non-decreasing order. Unlike selection sort which places only the minimum unsorted element in its correct position, this algorithm finds both the minimum and the maximum unsorted elements and places them into their final positions.

```
void  sort( list r[], int n )
{
   int i, j, mini, maxi;

   for (i=1; i<n-i+1; i++) {
      mini = maxi = i;
      for( j=i+1;                          (3分); ++j ){
         if(                               (3分) ) mini = j;
         else if(r[j]->key > r[maxi]->key) maxi = j;
      }
      if( mini != i ) swap(&r[mini], &r[i]);
      if( maxi != n-i+1 ){
         if(                               (3分) ) swap(&r[mini], &r[n-i+1]);
         else swap(&r[maxi], &r[n-i+1]);
      }
   }
}
```

5-2 The function is to find the `K`-th smallest element in a list `A` of `N` elements. The function `BuildMaxHeap(H, K)` is to arrange elements `H[1]` ... `H[K]` into a max-heap. Please complete the following program.

```
ElementType FindKthSmallest ( int A[], int N, int K )
{   /* it is assumed that K<=N */
    ElementType *H;
    int i, next, child;

    H = (ElementType *)malloc((K+1)*sizeof(ElementType));
    for ( i=1; i<=K; i++ ) H[i] = A[i-1];
    BuildMaxHeap(H, K);

    for ( next=K; next<N; next++ ) {
        H[0] = A[next];
        if ( H[0] < H[1] ) {
            for ( i=1; i*2<=K; i=child ) {
                child = i*2;
                if ( child!=K &&                          (3分) ) child++;
                if ( (3分) )
                    H[i] = H[child];
                else break;
            }
            H[i] = H[0];
        }
    }
    return H[1];
}
```

函数题

得分：暂无 总分：6

6-1 No Greater Than X in BST (6分)

You are supposed to output, in decreasing order, all the elements no greater than `X` in a binary search tree `T`.

Format of function:

```
void Print_NGT( Tree T,  int X );
```

where `Tree` is defined as the following:

```
typedef struct TreeNode *Tree;
struct TreeNode {
    int Element;
    Tree  Left;
    Tree  Right;
};
```

The function is supposed to use `Output(X)` to print `X`.

Sample program of judge:

```
#include <stdio.h>
#include <stdlib.h>

typedef struct TreeNode *Tree;
struct TreeNode {
    int Element;
    Tree  Left;
    Tree  Right;
};

Tree BuildTree(); /* details omitted */
void Output( int X ); /* details omitted */

void Print_NGT( Tree T,  int X );

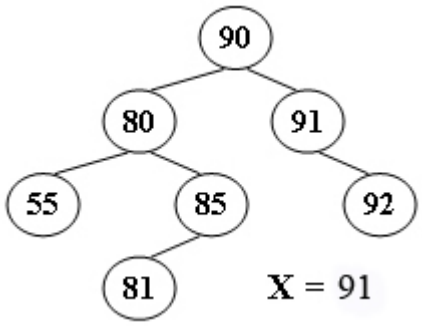
int main()
{
    Tree T;
    int X;

    T = BuildTree();
    scanf("%d", &X);
    Print_NGT( T, X );
    printf("End\n");

    return 0;
}

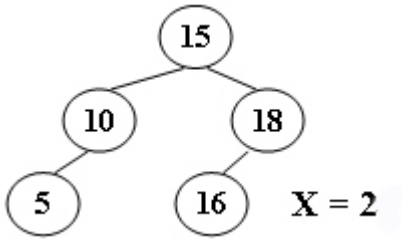
/* Your function will be put here */
```

Sample Output 1 (for the tree shown in Figure 1):




```
91 90 85 81 80 55 End
```

Sample Output 2 (for the tree shown in Figure 2):




```
End
```